

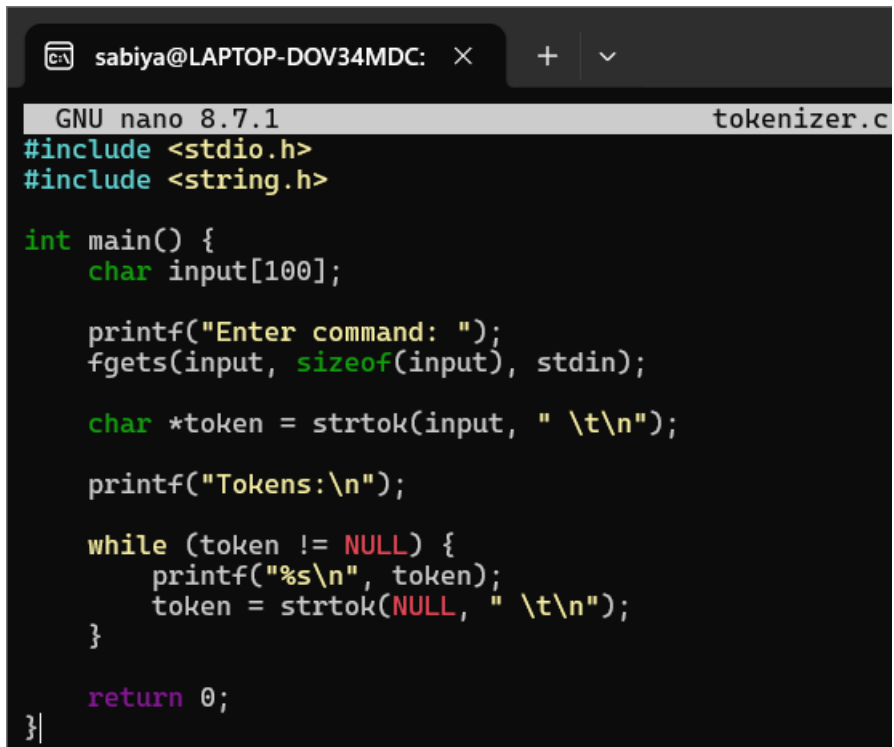
OSSP SKILL

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To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

(a) Code:

A screenshot of a terminal window with a dark background. The title bar shows 'sabiya@LAPTOP-DOV34MDC:'. The terminal content shows the GNU nano 8.7.1 editor editing 'tokenizer.c'. The code is as follows:

```
GNU nano 8.7.1 tokenizer.c
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

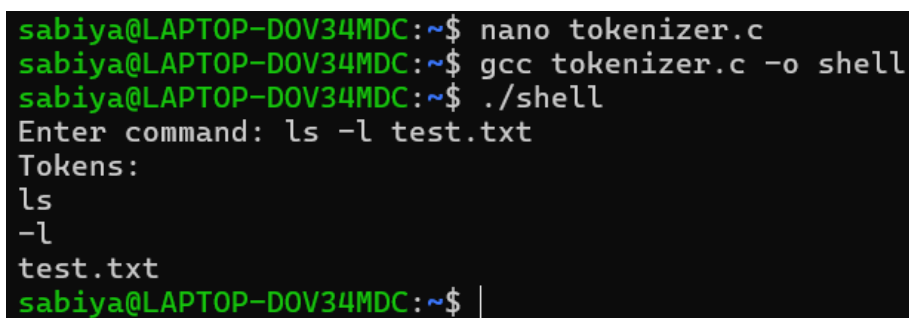
    char *token = strtok(input, " \\t\\n");

    printf("Tokens:\\n");

    while (token != NULL) {
        printf("%s\\n", token);
        token = strtok(NULL, " \\t\\n");
    }

    return 0;
}
```

OUTPUT SCREEN SHOT:

A screenshot of a terminal window showing the execution of the tokenizer program. The commands and output are as follows:

```
sabiya@LAPTOP-DOV34MDC:~$ nano tokenizer.c
sabiya@LAPTOP-DOV34MDC:~$ gcc tokenizer.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Enter command: ls -l test.txt
Tokens:
ls
-l
test.txt
sabiya@LAPTOP-DOV34MDC:~$ |
```

(b) Code:

```
GNU nano 8.7.1                                parser.c *
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0) {
        printf("Error: Empty command\n");
        return 0;
    }

    if (strcmp(input, "ls") == 0) {
        printf("Valid command: ls\n");
    }
    else if (strcmp(input, "pwd") == 0) {
        printf("Valid command: pwd\n");
    }
    else if (strcmp(input, "exit") == 0) {
        printf("Valid command: exit\n");
    }
    else {
        printf("Error: Invalid command\n");
    }

    return 0;
}
```

OUTPUT SCREENSHOT:

```
sabiya@LAPTOP-DOV34MDC:~$ nano parser.c
sabiya@LAPTOP-DOV34MDC:~$ gcc parser.c -o parser
sabiya@LAPTOP-DOV34MDC:~$ ./parser
Enter command: ls
Valid command: ls
sabiya@LAPTOP-DOV34MDC:~$ gcc parser.c -o parser
sabiya@LAPTOP-DOV34MDC:~$ ./parser
Enter command: hello
Error: Invalid command
sabiya@LAPTOP-DOV34MDC:~$ |
```